DFIR PLATFORM SPECIFICATION • SUTRAM 2026

READY FOR EXECUTION

AI-Powered Indigenous Digital Forensics Investigation Framework

System Architecture, Microservices Breakdown & 19-Step Forensic Workflow Implementation Plan

PROJECT TARGET	LEGAL COMPLIANCE	BLOCKCHAIN LEDGER	AI & ML MODELS
C:\My Projects\SUTRAM 2026\Prototype	Section 63 BSA 2023 / Part A & B	Polygon Amoy PoS (Chain ID 80002)	Gemini 2.0 Flash + Isolation Forest

Implementation Plan: AI-Powered Indigenous Digital Forensics Investigation Framework (DFIR)

Transform the baseline prototype in `C:\My Projects\SUTRAM 2026\Prototype` from a document locker into an active, intelligent **Digital Forensics and Incident Response (DFIR)** platform, fulfilling the challenge mandate and executing the complete **19-step investigative workflow**.

1. Workflow Architecture & Mapping

The platform will implement the end-to-end workflow deconstructed into 4 interconnected operational layers:

19-Step Master Forensic Workflow

LAYER 1: INGESTION & CUSTODY

1. Login & RBAC → 2. Create/Open Case → 3. Upload Evidence →
4. Hash + Register (Polygon) → 5. Process Evidence (Parsers) →
6. Normalize Artifacts

LAYER 2: EXTRACTION & CORRELATION

7. Extract Events & Entities → 8. Build Evidence Graph →
9. Correlate Sources → 10. Build Chronological Timeline

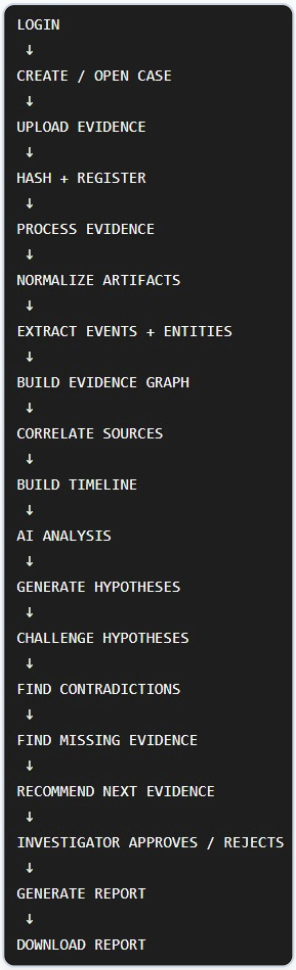
LAYER 3: AI FORENSIC REASONING

11. AI Analysis (Isolation Forest + NLP) → 12. Generate Hypotheses →
13. Challenge Hypotheses → 14. Find Contradictions →
15. Find Missing Evidence → 16. Recommend Next Evidence

LAYER 4: HUMAN-IN-THE-LOOP & STATUTORY COMPLIANCE

17. Investigator Approves/Rejects →
18. Generate Comprehensive Report →
19. Download Section 63 BSA Certificate

Reference Workflow Diagram



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2. User Review Required

> ⚠️ **CRITICAL REQUIREMENT**
>
> **Dependencies to Add:** To deliver Error Level Analysis (ELA), EXIF extraction, and ML Isolation Forest, we will add lightweight, battle-tested Python libraries to `backend/requirements.txt`: - `scikit-learn` (already installed for TF-IDF; will power `IsolationForest`) - `piexif` / `exifread` / `Pillow` (for EXIF metadata inspection) - `python-magic` or file signature detection for true MIME checking Frontend will integrate a lightweight canvas/SVG graph library (such as `@xyflow/react` or `vis-network` / HTML5 Canvas Force Graph) for the interactive **Detective Knowledge Graph**.

> ℹ️ **SYSTEM NOTE**
>
> **Original Prototype Safety:** All changes will occur exclusively in `C:\My Projects\SUTRAM 2026\Prototype`. The original directory `C:\My Projects\SIH_26190_Parikalp (2026)\Prototype` will remain 100% untouched.

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3. Proposed Changes

All paths below are relative to `C:\My Projects\SUTRAM 2026\Prototype`.

Backend: Core Forensic Engine & Microservices

NEW **parsers.py**

- **Format-Agnostic Ingestion Engine:**
- `parse_sqlite_db()`: Natively inspects uploaded `.db` / `.sqlite` files (e.g. mobile extractions, call logs, SMS, WhatsApp `msgstore.db`), querying tables and extracting message bodies, phone numbers, and timestamps.
- `parse_csv_logs()`: Parses firewall, Apache/Nginx, auth, and syslog CSV/TSV/JSON files, extracting IP addresses, HTTP status codes, usernames, and timestamped actions.
- `parse_chat_export()`: Ingests WhatsApp/Telegram `.txt` or `.json` exports, breaking them into message rows (Sender, Timestamp, Message content).

NEW **image_forgery.py**

- **Visual Forgery & Tamper Lab (Pillar 2):**
- **Error Level Analysis (ELA):** Resaves JPEG at a controlled quality rate (90-95%) and computes the absolute pixel difference scaled across color channels. Generates a visual ELA mask highlighting modified, spliced, or digitally drawn pixels (e.g., altered UPI IDs, edited receipts).
- **Noise Inconsistency Check:** Measures localized noise standard deviations across block tiles to detect copy-move/splicing forgery.
- Returns a Tamper Risk Score (0-100%) and an ELA visual diff overlay path.

NEW **metadata_forensics.py**

- **EXIF & Document Metadata Extractor (Pillar 2):**
- Strips and parses EXIF tags: Camera Make/Model, Software used (e.g., Photoshop, Canva, GIMP tags), Lens info, Original DateTime vs Digitized DateTime.
- Geolocation extraction: Parses GPS Latitude/Longitude and converts to human-readable decimal coordinates.
- PDF Metadata parser: Extracts Producer, Creator, CreationDate, ModDate, and flags discrepancies where Modification time precedes Creation time or points to pirated editing software.

NEW **log_anomaly.py**

- **Machine Learning Log Triage (Pillar 3):**
- `IsolationForest` model to detect outliers across access/firewall events based on:
  - Failed login count per unit time
  - Off-hours / abnormal time-of-day bursts
  - Rapid geographic dispersion (unusual foreign IP bursts)
  - High-frequency port sweeps / error responses (401, 403, 500)
- Returns flagged anomaly timestamps with anomaly scores and explanation tags.

NEW **correlation_engine.py**

- **Multi-Source Correlation & Event Extraction (Pillar 4):**
- **Entity Extractor (Regex + NER):** Identifies Phone Numbers (`+91` / 10-digit Indian formats), UPI IDs (`[a-zA-Z0-9.\-_]+@[a-zA-Z]+`), IP Addresses (IPv4/IPv6), Email IDs, Crypto Wallets, Bank Accounts, Hashes, and Person names.
- **Cross-Source Correlator:** Maps intersections where an entity appears across different exhibits (e.g., Phone in WhatsApp chat $\cap$ UPI ID in screenshot $\cap$ Destination IP in firewall log).

- **Master Event Normalizer:** Converts disparate timestamps into unified UTC events with standard severity levels (`INFO`, `SUSPICIOUS`, `CRITICAL`).

NEW **investigative_ai.py**

- **Hypothesis, Contradiction & Lead Recommendation Engine (Steps 12 - 16):**
- Integrates Gemini 2.0 Flash to reason over the correlated evidence matrix:
  1. **Generate Hypotheses:** Synthesizes 2-3 plausible narratives of how the crime unfolded (Modus Operandi, suspect vectors, entry points).
  2. **Challenge Hypotheses:** Acts as an adversarial "Devil's Advocate" testing alternate explanations (e.g., credential stuffing vs insider breach).
  3. **Find Contradictions:** Flags incompatible timestamps, impossible travel speeds between IP and GPS, or conflicting alibis.
  4. **Detect Missing Evidence:** Identifies evidentiary voids needed for trial (e.g., missing ISP logs, CDR records, bank statements).
  5. **Recommend Next Steps:** Suggests specific legal/forensic actions (e.g., "Issue notice under Section 94 BNSS to telecom provider for CDR").

NEW **bsa_certificate.py**

- **Statutory Compliance Certificate (Section 63 BSA 2023):**
- Implements the legally structured statutory format for electronic evidence admissibility in Indian courts:
  - **Part A:** Identifying particulars of the electronic record, source device specifications, extraction tools, and cryptographic hashes (SHA-256 un-salted & blockchain proof).
  - **Part B:** Statutory declaration by the officer in charge / technical examiner certifying electronic custody integrity and absence of tampering.

MODIFY **models.py**

- Extend schema to support:
- `EvidenceArtifact`: Generic evidence container storing artifact category (`DOCUMENT`, `MOBILE_DB`, `NETWORK_LOG`, `CHAT_EXPORT`, `MEDIA`), raw SHA-256, salted SHA-256, metadata JSON (EXIF, GPS, device), and forgery analysis JSON (ELA score).
- `ForensicEvent`: Normalized event stream (`case_id`, `artifact_id`, `timestamp`, `event_type`, `description`, `severity`, `source`).
- `ForensicEntity`: Extracted entities (`entity_type`, `value`, `case_id`, `occurrences`).
- `EntityLink`: Graph edge representing relationships between entities (`source_id`, `target_id`, `relation_type`, `confidence`).
- `InvestigationHypothesis`: Hypotheses (`case_id`, `title`, `description`, `supporting_evidence`, `counter_evidence`, `status`: `'Proposed'`, `'Approved'`, `'Rejected'`, `notes`).
- `InvestigationLead`: Actionable leads and missing evidence recommendations with IO approval status.

MODIFY **main.py**

- Add API endpoints:
- `POST /cases/{id}/artifacts/`: Multi-format evidence ingestion pipeline.
- `GET /artifacts/{id}/forgery/`: Returns ELA image diff overlay and tamper analysis.
- `GET /artifacts/{id}/metadata/`: Returns EXIF, GPS, and software tags.

- `GET /cases/{id}/timeline/`: Returns unified normalized chronological timeline.
- `GET /cases/{id}/graph/`: Returns nodes and edges for the evidence relationship graph.
- `GET /cases/{id}/anomalies/`: Triggers ML log anomaly triage and returns flagged points.
- `POST /cases/{id}/hypotheses/generate/`: Runs AI investigative reasoning.
- `POST /hypotheses/{id}/review/`: Allows investigator to Approve, Reject, or annotate hypotheses.
- `GET /cases/{id}/bsa-certificate/`: Generates statutory Section 63 BSA digital certificate.

Frontend: Interactive Forensic Investigation Portal

MODIFY **case/[id]/page.tsx**

- Transform the single-purpose document view into a multi-tab **Forensic Command Center**:
- **Tab 1: Evidence Locker & Artifact Ingestion:** Upload any artifact type, preview files, trigger ELA inspection, view EXIF/GPS badges, and run live Polygon tamper verification.
- **Tab 2: Master Incident Timeline:** Visual interactive chronological timeline showing cross-source events, filtering by severity (`CRITICAL` , `WARNING` , `INFO`), actors, and artifact types.
- **Tab 3: Entity Knowledge Graph (Detective Board):** Node-and-edge visual canvas linking phone numbers, UPI IDs, IPs, and names across exhibits with zoom/pan and cluster inspection.
- **Tab 4: AI Forensic Lab & Hypotheses:** Displays ML log anomaly scores, AI hypotheses with adversarial counter-arguments, detected contradictions, and missing evidence alerts. Includes IO **Approve / Reject** buttons.
- **Tab 5: Statutory Reports & BSA Certificate:** Live preview and download of the official **Section 63 BSA 2023 Statutory Certificate** and full incident dossier.

NEW **components/EvidenceGraph.tsx**

- Interactive canvas/SVG knowledge graph component:
- Custom node badges (color-coded: Phone = Orange, UPI = Green, IP = Blue, Person = Purple, File = Gray).
- Hover tooltip displaying occurrences and linked evidence files.
- Click-to-highlight connected sub-graphs.

NEW **components/MasterTimeline.tsx**

- Chronological incident reconstruction stream:
- Time-ordered cards with icons for network events, chat messages, financial transfers, and file modifications.
- Anomaly markers for events flagged by ML Isolation Forest.

NEW **components/ForgeryModal.tsx**

- ELA & Metadata Inspection Modal:
- Side-by-side or slider comparison of original image vs Error Level Analysis (ELA) compression heatmap.
- EXIF metadata table with GPS map coordinate link and software editing history.

NEW **components/BSACertificateModal.tsx**

- Printable, government-formatted modal rendering Section 63 BSA Part A & Part B electronic certificate ready for court submission.

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4. Verification Plan

Automated & Unit Tests

1. **Parser Tests (`test_parsers.py`):**
2. Test SQLite parser on sample mobile DB.
3. Test CSV log parser on sample Apache/firewall access logs.
4. Test WhatsApp text chat parser.
5. **Forensic Analysis Tests (`test_forensics.py`):**
6. Test ELA generation on genuine vs altered image (e.g. modified transaction receipt).
7. Test EXIF extractor on sample photo with GPS coordinates.
8. Test ML `IsolationForest` on normal vs burst login access dataset.
9. **Correlation & Graph Tests (`test_correlation.py`):**
10. Verify phone number in chat links with identical UPI ID and IP address into graph nodes and edges.
11. **BSA Compliance Verification (`test_bsa.py`):**
12. Verify output matches Section 63 BSA statutory format containing required device specs and un-salted SHA-256.

Manual Verification

1. Log in as Sub-Inspector Sharma (`sharma` / `Officer@123`).
2. Create new Cyber Incident case.
3. Ingest heterogeneous evidence:
4. Sample altered UPI screenshot $\rightarrow$ Open Forgery Lab $\rightarrow$ observe ELA heat map flagging edited digits.
5. Sample server log CSV $\rightarrow$ Observe ML Isolation Forest detecting brute force spike.
6. Sample WhatsApp chat $\rightarrow$ Observe entities extracted.
7. Open **Evidence Graph** $\rightarrow$ Verify cross-linking between phone, UPI, and IP.
8. Open **Master Timeline** $\rightarrow$ Verify synchronized incident timeline.
9. Open **AI Hypotheses** $\rightarrow$ Review generated hypotheses, contradictions, and missing evidence recommendations $\rightarrow$ Click **Approve**.
10. Download **Section 63 BSA Statutory Certificate** and verify court-ready formatting.